

Zheda Mai

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EDUCATION	Ph.D. Computer Science and Engineering, Ohio State University 2022-2027 (Expected) - Research areas: Efficient Robust Interpretable Model Adaptation, Vision Foundation Models, Multimodal LLMs, Continual Learning, Robotics - Advisor: Professor Wei-Lun Chao.
	M.A.Sc. Information Engineering, University of Toronto 2018-2021 - Research areas: Continual Learning, Recommender Systems - Advisor: Professor Scott Sanner.
	B.A.Sc. Engineering Science, University of Toronto 2012-2017
RESEARCH EXPERIENCE	Research Intern, Amazon Fauna Robotics May 2026 - Aug. 2026 Vision-Language-Action model adaptation.
	Research Intern, Amazon Lab126 May 2025 - Aug. 2025 Efficient MLLMs with model stitching.
	Research Intern, Bosch Research May 2024 - Aug. 2024 Time series analysis with MLLMs.

PUBLICATIONS *: Co-First Authorship. †: Project Lead.

Conferences

- [ECCV'26] Sooyoung Jeon, Hongjie Tian, Lemeng Wang, **Zheda Mai**[†], Vidhi Bakshi, Jiacheng Hou, Ping Zhang, Arpita Chowdhury, Wei-Lun Chao. Lessons and Open Questions from a Unified Study of Camera-Trap Species Recognition Over Time. *ECCV*, 2026.
- [CVPR'26] **Zheda Mai**, Ke Zhang, Fu-En Wang, Zixiao Ken Wang, Albert Y. C. Chen, Lu Xia, Min Sun, Wei-Lun Chao, Cheng-Hao Kuo. Revisiting Model Stitching in the Foundation Model Era. *CVPR*, 2026.
- [CVPR'26] **Zheda Mai**^{*}, Arpita Chowdhury^{*}, Zihe Wang, Sooyoung Jeon, Lemeng Wang, Jiacheng Hou, Jihyung Kil, Wei-Lun Chao. AVA-Bench: Atomically Visual Ability Benchmark for Vision Foundation Models. *CVPR*, 2026.
- [ICLR'26] **Zheda Mai**^{*}, Justin Lee^{*}, Chongyu Fan, Wei-Lun Chao. An Empirical Exploration of Continual Unlearning for Image Generation. *ICLR*, 2026.
- [NeurIPS'25] **Zheda Mai**^{*}, Ping Zhang^{*}, Quang-Huy Nguyen, Wei-Lun Chao. Revisiting Semi-Supervised Learning in the Era of Foundation Models. *NeurIPS*, 2025.
- [NeurIPS'25] Jianyang Gu, Samuel Stevens, Elizabeth G Campolongo, Matthew J Thompson, Net Zhang, Jiaman Wu, Andrei Kopanav, **Zheda Mai**, Alexander E. White, James Balhoff, Wasla Dahdul, Daniel Rubenstein, Hilmar Lapp, Tanya Berger-Wolf, Wei-Lun Chao, Yu Su. Bioclip 2: Emergent properties from scaling hierarchical contrastive learning. *NeurIPS*, 2025 (**Spotlight**).
- [CVPR'25] **Zheda Mai**, Ping Zhang, Cheng-Hao Tu, Hong-You Chen, Li Zhang, Wei-Lun Chao. Lessons and Insights from a Unifying Study of Parameter-Efficient Fine-Tuning (PEFT) in Visual Recognition. *CVPR*, 2025 (**Highlight**).
- [CVPR'25] Arpita Chowdhury, Dipanjoy Paul, **Zheda Mai**, Jianyang Gu, Ziheng Zhang, Kazi Sajeed Mehrab, Elizabeth G Campolongo, Daniel Rubenstein, Charles V Stewart, Anuj Karpatne, Tanya Berger-Wolf, Yu Su, Wei-Lun Chao. Prompt-CAM: Making Vision Transformers Interpretable for Fine-Grained Analysis. *CVPR*, 2025.
- [CVPR'25] Ziheng Zhang, Jianyang Gu, Arpita Chowdhury, **Zheda Mai**, David Carlyn, Tanya Berger-Wolf, Yu Su, Wei-Lun Chao. Finer-CAM: Spotting the Difference Reveals Finer Details for Visual Explanation. *CVPR*, 2025.

- [NAACL’25] Zhongwei Wan, Hui Shen, Xin Wang, Che Liu, **Zheda Mai**, Mi Zhang. Attention Entropy-Guided Dynamic Cache Allocation for Efficient Multimodal Long-Context Inference. *NAACL*, 2025.
- [ICASSP’25] Jiageng Zhu, Kehao Li, **Zheda Mai**, Hanchen Xie, Wael AbdAlmageed, Zubin Abraham. Attention-Driven Causal Discovery: From Transformer Matrices to Granger Causal Graphs for Non-Stationary Time-series Data. *ICASSP*, 2025.
- [NeurIPS’24] **Zheda Mai**^{*}, Jihyung Kil^{*}, Justin Lee, Zihe Wang, Kerrie Cheng, Lemeng Wang, Ye Liu, Arpita Chowdhury, Wei-Lun Chao. MLLM-CompBench: A Comparative Reasoning Benchmark for Multimodal LLMs. *NeurIPS*, 2024.
- [NeurIPS’24] **Zheda Mai**^{*}, Arpita Chowdhury^{*}, Ping Zhang^{*}, Cheng-Hao Tu, Hong-You Chen, Vardaan Pahuja, Tanya Berger-Wolf, Song Gao, Charles Steward, Yu Su, Wei-Lun Chao. Fine-Tuning is Fine, if Calibrated. *NeurIPS*, 2024.
- [NeurIPS’23] Cheng-Hao Tu^{*}, Hong-You Chen^{*}, **Zheda Mai**, Jike Zhong, Vardaan Pahuja, Tanya Berger-Wolf, Song Gao, Charles Steward, Yu Su, Wei-Lun Chao. Holistic Transfer: Towards Non-Disruptive Fine-Tuning with Partial Target Data. *NeurIPS*, 2023.
- [CVPR’23] **Zheda Mai**^{*}, Cheng-Hao Tu^{*}, Wei-Lun Chao. Visual Query Tuning: Towards Effective Usage of Intermediate Representations for Parameter and Memory Efficient Transfer Learning. *CVPR*, 2023.
- [WWW’22] **Zheda Mai**^{*}, Tianshu Shen^{*}, Ga Wu, Scott Sanner. Distributional Contrastive Embedding for Clarification-based Conversational Critiquing. *WWW*, 2022.
- [SIGIR’22] Zhaolin Gao, Tianshu Shen, **Zheda Mai**, Mohamed Reda Bouadjenek, Scott Sanner. Mitigating the Filter Bubble while Maintaining Relevance: Targeted Diversification with VAE-based Recommender Systems. *SIGIR*, 2022.
- [AAAI’21] **Zheda Mai**^{*}, Dongsub Shim^{*}, Jihwan Jeong^{*}, Scott Sanner, Hyunwoo Kim, Jongseong Jang. Online Class-Incremental Continual Learning with Adversarial Shapley Value. *AAAI*, 2021 (**Oral**).

Journals

- [TMLR’26] Qinghua Liu^{*}, Sam Heshmati^{*}, **Zheda Mai**^{*}, Zubin Abraham, John Paparrizos, Liu Ren. MLLM4TS: Leveraging Vision and Multimodal Language Models for General Time-Series Analysis. *TMLR*, 2026.
- [TMLR’26] Xi Xiao, Yunbei Zhang, Lin Zhao, Yiyang Liu, Xiaoying Liao, **Zheda Mai**, Xingjian Li, Xiao Wang, Hao Xu, Jihun Hamm, Xue Lin, Min Xu, Qifan Wang, Tianyang Wang, Cheng Han. Prompt-based Adaptation in Large-scale Vision Models: A Survey. *TMLR*, 2026.
- [JVCIR’23] Ruiwen Li, **Zheda Mai**, Chiheb Trabelsi, Zhibo Zhang, Jongseong Jang, Scott Sanner. TransCAM: Transformer Attention-based CAM Refinement for Weakly Supervised Semantic Segmentation. *JVCIR*, 2023.
- [IPM’23] Tianshu Shen, Jiaru Li, Mohamed Reda Bouadjenek, **Zheda Mai**, Scott Sanner. Unintended Bias in Language Model-driven Conversational Recommendation. *IPM*, 2023.
- [Neurocomputing’22] **Zheda Mai**, Ruiwen Li, Jihwan Jeong, David Quispe, Hyunwoo Kim, Scott Sanner. Online Continual Learning in Image Classification: An Empirical Survey. *Neurocomputing*, 2022.
- [AIJ’22] Vincenzo Lomonaco, Lorenzo Pellegrini, Pau Rodriguez, Massimo Caccia, Qi She, Yu Chen, Quentin Jodelet, Ruiping Wang, **Zheda Mai**, David Vazquez, German I Parisi, Nikhil Churamani, Marc Pickett, Issam Laradji, Davide Maltoni. CVPR 2020 Continual Learning in Computer Vision Competition: Approaches, Results, Current Challenges and Future Directions. *AIJ*, 2022.

Workshops

- [IROS’26] Sooyoung Jeon, Zhen Xu, Jae Young Lim, Jinsu Yoo, Yunbei Zhang, Jiacheng Hou, **Zheda Mai**[†], Wei-Lun Chao. Do All Skills Forget Alike? A Preliminary Study of Task-Specific Forgetting in Continual Vision-Language-Action Models. *IROS*, 2026.
- [IROS’26] Sooyoung Jeon, Zhen Xu, Jae Young Lim, Jinsu Yoo, Yunbei Zhang, Jiacheng Hou, **Zheda Mai**[†], Wei-Lun Chao. Benchmarking Heterogeneous and Compositional Continual Learning for Vision-Language-Action Models. *IROS*, 2026.
- [CVPR’26] Lemeng Wang, Qinqian Lei, Vidhi Bakshi, Daniel Yi, Yifan Liu, Jiacheng Hou, Asher Seng Hao,

Zheda Mai, Wei-Lun Chao, Robby T Tan, Bo Wang. A Study of Failure Modes in Two-Stage Human-Object Interaction Detection. *CVPR*, 2026.

[NeurIPS'25] Sooyoung Jeon, **Zheda Mai**, Hongjie Tian, Vidhi Bakshi, Lemeng Wang, Jiacheng Hou, Ping Zhang, Arpita Chowdhury, Wei-Lun Chao. Continually Adapt or Not (CAN)? A Continual Learning Benchmark of Camera Trap Species Classification over Time. *NeurIPS*, 2025.

[CVPR'25] Hanchen Xie, Rose Ma, Jiageng Zhu, **Zheda Mai**, Wael Abd-Almageed, Zubin Abraham. Mitigating Video Content Misalignment on Large Vision Model with Time-Series Data Alignment. *CVPR*, 2025

[NeurIPS'23] **Zheda Mai***, Tianle Chen*, Ruiwen Li, Wei-Lun Chao. Segment Anything Model (SAM) Enhanced Pseudo Labels for Weakly Supervised Semantic Segmentation. *NeurIPS*, 2023.

[CVPR'21] **Zheda Mai**, Ruiwen Li, Hyunwoo Kim, Scott Sanner. Supervised Contrastive Replay: Revisiting the Nearest Class Mean Classifier in Online Class-Incremental Continual Learning. *CVPR*, 2021.

[CVPR'20] **Zheda Mai**, Hyunwoo Kim, Jihwan Jeong, Scott Sanner. Batch-level Experience Replay with Review for Continual Learning. *CVPR*, 2020.

[ICDM'20] **Zheda Mai***, Ga Wu*, Kai Luo, Scott Sanner. Attentive Autoencoders for Multifaceted Preference Learning in One-class Collaborative Filtering. *ICDM*, 2020.

Under Review

[R1] **Zheda Mai**, Sooyoung Jeon, Zanming Huang, Jinsu Yoo, Colin Lee, Gengyu Zhang, Haoxuan Wang, Yifan Li, Yu Kong, Yan Yan, Wei-Lun Chao. A Survey of Continual Learning for Robotics in the Foundation Model Era, 2025

[R2] Yifan Li, Lichi Li, Anh Dao, Xinyu Zhou, Yicheng Qiao, **Zheda Mai**, Daeun Lee, Zichen Chen, Zhen Tan, Mohit Bansal, Yu Kong. IndustryNav: Exploring Spatial Reasoning of Embodied Agents in Dynamic Industrial Navigation

FUNDING

- National Artificial Intelligence Research Resource (**NAIRR**): A Comprehensive Understanding of Continual Learning for Vision-Language-Action Models (Project lead). 20,000 GPU hours (2026-2027).
- Meta Aria-Gen2 glasses (Project lead). 2026.

AWARDS

OSU Graduate Student Research Award, 2026
OSU Graduate Student Travel Award, 2026
Outstanding Reviewer Award for **NeurIPS**, 2023
4th place of the CLVision Continual Learning challenge at **CVPR**, 2021
1st place of the CLVision Continual Learning challenge at **CVPR**, 2020
NEXT36 Fellow from NEXT Canada, 2018

PROFESSIONAL SERVICE

I am the organizer and chair for

- NeurIPS'25 Imageomics Workshop
- WACV'26 Human-Centric Open-World Understanding and Reasoning Workshop

I am a reviewer for

- NeurIPS: 2023, 2024, 2025
- ICML: 2023, 2024
- ICLR: 2024, 2025
- CVPR: 2024, 2025, 2026
- ECCV: 2026
- WACV: 2026
- IJCAI: 2024
- Transactions on Pattern Analysis and Machine Intelligence (TPAMI)
- ACM Computing Surveys
- Artificial Intelligence (AIJ)
- Frontiers in Artificial Intelligence

MENTORING	Sooyoung Jeon (OSU M.S. → BU Ph.D.; 2026-2027): Continual Learning for Robotics Justin Lee (OSU B.S. → UMich Ph.D.; 2024-2026): Continual Machine Unlearning Lemeng Wang (OSU B.S. → Stanford M.S.; 2024-2026): Foundation Model Evaluation Hongjie Tian (OSU B.S. → UCF Ph.D; 2025): Efficient Adaptation Tianle Chen (OSU B.S. → BU Ph.D.; 2022-2023): Weakly Supervised Semantic Segmentation Tianshu Shen (UToronto M.S. 2021-2022): Conversational Recommender Systems Ruiwen Li (UToronto M.S. 2020-2022): Weakly Supervised Semantic Segmentation
OTHER EXPERIENCE	Data Scientist, <i>Optimy AI</i>, Canada Jan. 2021-Aug. 2022 - Developed machine learning models for customer engagement and purchase predictions. Machine Learning Engineer Intern, <i>Pitney Bowes</i>, Canada May 2019 - Oct. 2019 - Developed map-style extraction models with CNN and multi-task learning. Software Engineer Intern, <i>AMD</i>, Canada May 2015 - May 2016 - Developed design verification tool automation for Verilog.
TEACHING	Teaching Assistant, Ohio State University - CSE5523: Machine Learning and Statistical Pattern Recognition (2026) - CSE5524: Foundations of Computer Vision (2025) Teaching Assistant, University of Toronto - APS1070: Foundations of Data Analytics and Machine Learning (2019, 2020) - MIE451: Decision Support Systems (2019, 2020) - MIE1628: Big Data Science (2020)
VOLUNTEER	Judge and mentor OSU Artificial Intelligence Hackathon: 2022, 2023, 2024